

Time-Dependent Control of Voltage and Current via a Tunable Metasurface

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Abstract—Metasurfaces have recently attracted significant attention due to their potential use in optics and meta-devices as compact platforms for precise manipulation of electromagnetic waves. In this work, we present a tunable metasurface geometry based on a triple concentric ring configuration that offers superior surface-wave confinement and enhanced reactive control compared with conventional single-ring or patch structures. Based on our designs, the proposed structure achieves controllable voltage and current responses through strong field confinement and rapid impedance modulation along its surface. Although implementation on a cost-effective FR-4 substrate exhibits moderate dielectric loss at high frequencies, the metasurface maintains efficient performance and supports surface-wave transmission at 30 GHz. Under sinusoidal excitation with an input amplitude of 1 V at 30 GHz, the device outperforms a copper reference plate, providing approximately 40 V of voltage and about 0.8 A of current at the microscale across a resistive load. Temporal reconfiguration is enabled by an exponential voltage signal applied to the voltage-controlled switches (VCS), which alternates the surface reactance between capacitive and inductive states within a 20 ns window. The electromagnetic field amplitudes remain stable across the metasurface, and the switching process produces controlled suppression and restoration of power flow. Overall, the metasurface demonstrates tunable, time-dependent electrical control and holds promise for next-generation reconfigurable communication, sensing, and energy-modulation platforms.

Index Terms—Metasurface, Voltage Controlled Switch, Temporal modulation, Surface reactance, Temporal boundaries, FR-4 substrate.

I. INTRODUCTION

Electromagnetic media may undergo abrupt changes either in space or time. Spatially, material properties change across an interface at a boundary, while in time, the material undergoes an abrupt change in its properties at a given instant without variation in space [1]. Temporal discontinuities are special interfaces that do not conserve energy in the usual sense but conserve momentum instead. Consequently, waves impinging on such interfaces experience shifts of angular frequency rather than simple reflections or transmissions [2]. The foundation for temporal discontinuities has been developed theoretically over time, while experiments have confirmed their effects across various frequency regimes: from frequency mixing and wave manipulation in terahertz systems [3] and

optical setups [4] to microwave demonstrations [5]. These results open perspectives for employing time-varying discontinuities as a versatile means to manipulate waves over very broad spectra. Metasurfaces, two-dimensional arrays of sub wavelength scatterers, already allow impressive control over reflection, transmission, and polarization [6]. Incorporating temporal discontinuities into these structures can extend their functionality to enable dynamic modulation of surface waves in ways that static metasurfaces cannot achieve.

In this work, we examine how abrupt transitions in the effective surface impedance of a metasurface impact the voltage and current delivered into a resistive load. Switching the boundary condition from a capacitive state to an inductive state suppresses the induced current and voltage, while reverting to the capacitive state rapidly restores both quantities. This dynamic behavior provides a level of tunability unattainable with conventional metallic surfaces and enables new mechanisms for reconfigurable electromagnetic control. This paper makes three main contributions: At first, the paper propose a triple-ring metasurface that enhances surface-wave confinement and improves electromagnetic coupling along the structure. Secondly, it demonstrates a temporal switching mechanism capable of freezing and subsequently restoring electromagnetic fields through abrupt impedance transitions. Finally, We achieve regulated voltage and current delivery to a load by exploiting time-varying surface impedance, enabling controllable power transfer across the metasurface.

II. LITERATURE REVIEW

The earliest theoretical treatments of temporal boundaries, such as those by Morgenthaler [7], established that time-varying materials enforce momentum conservation and induce frequency changes in propagating waves. Later studies expanded these ideas using time-transformation frameworks to better describe reflection and transmission at abrupt temporal interfaces [6]. Experimental validation has since confirmed that frequency conversion caused by temporal discontinuities can be realized in practical terahertz [4], optical [5], and microwave [6] settings. Research has also explored how temporal modulation affects different classes of materials. Work on temporal slabs [8], anisotropic media [9], chiral materials [10],

and magneto-plasmas [11] has revealed unusual effects including non reciprocity, beam splitting, polarization conversion, and frequency mixing. Further efforts have extended these concepts to structured systems, including temporal coatings designed to suppress reflection [12], cavities and leaky waveguides supporting time-dependent field confinement [13], [14], graphene-based surfaces exhibiting tunable temporal responses [15], and polarization splitting [16].

Within the field of engineered metasurfaces, the introduction of time-varying control has gained momentum only recently. Early reviews on spatiotemporally modulated metasurfaces demonstrated their ability to dynamically manipulate light fields [17], while later work highlighted broader opportunities in photonic metamaterials [18]. Wang et al. [19] developed a rigorous theoretical model of temporal discontinuities in reactive metasurfaces, showing how surface waves can be slowed, halted, or redirected through fast impedance switching. Related studies demonstrated that temporal modulation could couple free-space and surface waves, acting as a temporal analog of a prism [19]. Other contributions have focused on practical realizations. Coherently modulated metasurfaces were introduced to enable multi-band absorption and active control of electromagnetic energy flow [1]. More recently, passive waveform-selective metasurfaces capable of responding differently to pulses and continuous waves without external bias have been reported [11]. Together, these works demonstrate that introducing a temporal degree of freedom into metasurfaces produces a wide range of novel electromagnetic phenomena. Concurrently, chiral hyperbolic metamaterials have been used to generate strong directional forces at the nanoscale, making it possible to manipulate and sort nanorods effectively [20]. In addition, multilayer dielectric biosensor designs have demonstrated tight field confinement and high refractive-index sensitivity [21]–[23]. Together, these results show the growing potential of temporal and metamaterial structures for advanced electromagnetic and optical control.

Despite this growing body of research, the direct use of temporal discontinuities to control electrical outputs such as voltage and current at a load has not been thoroughly examined. This paper aims to address that gap by analyzing surface-wave-driven metasurfaces with rapidly varying impedances, thereby extending temporal boundary theory into practical electrical control applications.

III. METHODOLOGY

In this study, a horn-type waveguide antenna is employed as the primary source to illuminate the designed metasurface with a Y -polarized electromagnetic wave. The incident beam strikes the metasurface at an oblique angle of 30° , as illustrated in Fig. 2, where Fig. 1 shows the dimensions of the unit cell of the metasurface consisting of 9 such unit cells. The metasurface was excited at 30 GHz using a horn antenna, linear polarization, and 30° incidence, with an input voltage of 1 V. Dynamic modulation was achieved through a voltage-controlled switch circuit, allowing real-time tuning of the surface response.

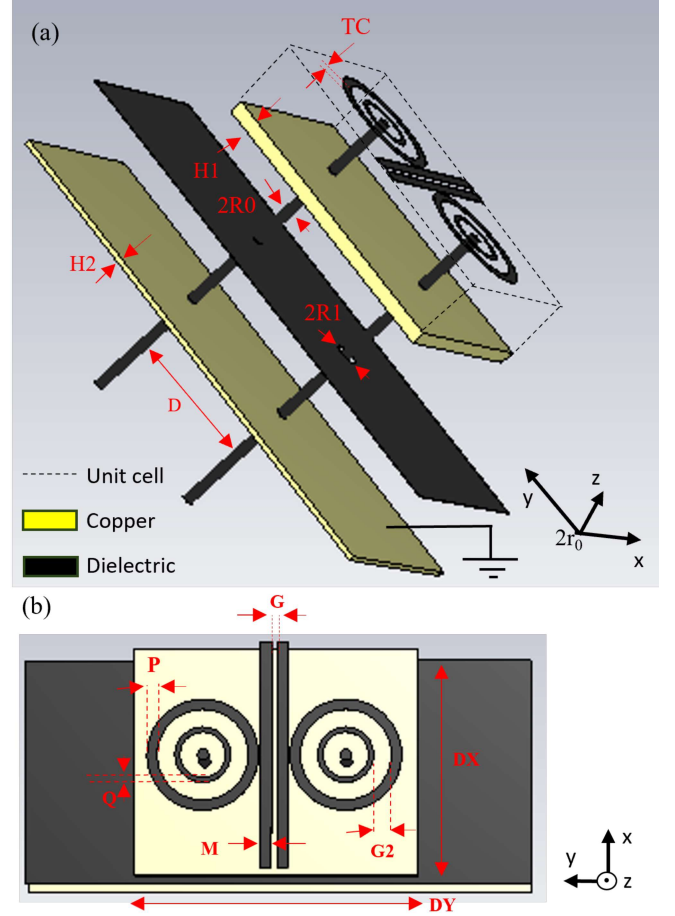


Fig. 1: The unit cell geometry is illustrated in multiple perspectives. Figure (a) presents the three-dimensional structure, while Figure (b) shows the two-dimensional projection on the xy -plane (top view). The corresponding geometric parameters are labeled in their respective positions: gap $G = 0.0678$ mm, inner diameter $2R0 = 0.1356$ mm, outer diameter $2R1 = 0.2712$ mm, conductor thickness $TC = 0.00678$ mm, dielectric height $H1 = 0.16272$ mm, total dielectric thickness $H2 = TC + 0.06102$ mm, substrate thickness $D = 1.55941$ mm, periodicity along y -axis $DY = 3.39$ mm, periodicity along x -axis $DX = 2.712$ mm, parameter $P = 0.12204$ mm, parameter $Q = 0.06102$ mm, spacing $M = 0.1356$ mm, and secondary gap $G2 = 0.21018$ mm.

The purpose of this arrangement is to evaluate the portion of the transmitted wave that successfully couples into the resistive load and propagates across the metasurface. These results are then compared against those obtained from a conventional copper plate of identical size (length, width, and height), serving as a reference structure. The overall schematic of the system is depicted in Fig. 2, where the metasurface is aligned along the longitudinal axis. A detailed circuit-level representation is provided in Fig. 3. Each unit cell of the metasurface incorporates three voltage-controlled switches

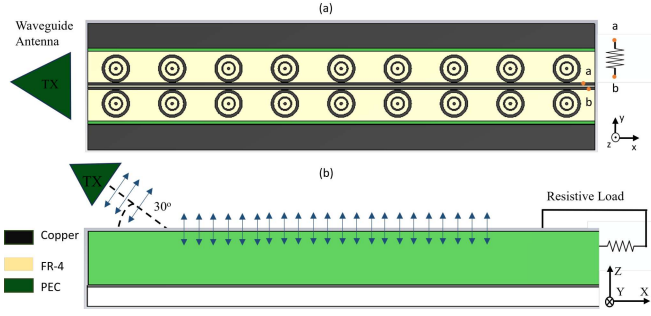


Fig. 2: The configuration places a waveguide antenna at the left boundary of the metasurface, while a resistive load is connected at the opposite boundary between two designated nodes, A and B, on the copper plate to measure the voltage and current across it. Figure (a) illustrates the arrangement in the xy -plane, showing nine unit cells distributed along the x -axis. Figure (b) depicts the zx -plane of the structure, where a horn waveguide antenna excites the metasurface with a y -polarized electromagnetic wave incident at a 30° angle. .

(VCSs), which enable dynamic reconfiguration by alternating between short-circuit and open-circuit states across the paired copper plates of the structure. This configuration establishes a controllable temporal boundary within the surface. To induce

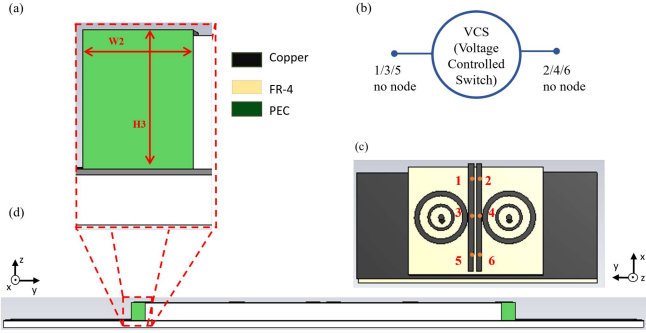


Fig. 3: An operational schematic illustrating the temporal functionality within a unit cell is presented. Figure (c) illustrates the unit cell that incorporates three voltage-controlled switches (VCS). The left copper layer is connected to nodes labeled 1, 3, and 5, while the right layer corresponds to nodes 2, 4, and 6. Figure (b) highlights a single VCS with two terminals. In Figure (d), a cross-sectional view of the unit cell is provided, displaying the PEC blocks positioned adjacent to the substrate along the x -axis, while Figure (a) offers a clearer view with annotated dimensions. The height of the PEC block is given as $H3 = H1 + TC = 0.1695$ mm, and its width is $W2 = 0.1356$ mm.

temporal modulation, an exponential control signal is applied to the VCS network. Under normal operation, the switches remain open; however, between $t = 7$ ns and $t = 17$ ns, the switches momentarily form short-circuit connections across the plates, thereby activating the temporal effect. The active

switching nodes are denoted as 1, 2, 3, 4, 5, and 6, which define the critical coupling points within the circuit.

Additionally, two perfectly electric conductor (PEC) blocks are strategically positioned on either side of the metasurface along the y -axis, as shown in Fig. 3(a,d). These act as shielding boundaries, ensuring that the electromagnetic energy is guided efficiently from the horn source to the load. The total simulation time is set to 20 ns, allowing the transient effects of the temporal modulation to be fully captured and analyzed. There is an additional observation to observe how much free space can reach the resistive load, also that sustains up to the end edge of the metasurface, compared to the conventional copper plate having the same dimensions (length, width, height).

IV. RESULTS AND DISCUSSION

This section examines the controlled modulation of power along a metasurface, characterized by the combined voltage and current passed across a resistive load. The source and the load are placed at opposite ends along the metasurface's length. A Y -polarized electric field, representing a sinusoidal electromagnetic wave at 30 GHz, is incident at a 30° angle within the xz -plane. The wave is generated using a waveguide antenna, with the full structural and circuit details described in the methodology section. By precisely regulating the voltage and current signals, we observe that energy sufficiently reaches and passed across the resistive load, as shown in Fig. 4. For the designed metasurface, however, the power is intentionally suppressed to zero between 7 ns and 17 ns, while propagation begins at 4 ns, as illustrated in Fig. 4(a, b). The boundary reactance transitions from an initial capacitive state to an inductive state and back to a capacitive state ($C_1 \rightarrow L \rightarrow C_2$), as shown in Fig. 6(d). When the reactance shifts to the inductive state, both the voltage and the current at the load drop to zero. During this interval, the static magnetic field along the boundary becomes frozen, the surface wave disappears entirely, and the electric field vanishes. When the boundary returns to a capacitive state, current resumes flowing through the load, while the voltage stabilizes. This effect is achieved by actively modifying the boundary condition of the metasurface, starting with the incident TE-polarized wave defined as:

$$E_i = E_0 \sin(10\pi x + 10\pi\sqrt{3}z - 60\pi \times 10^9 t) \hat{y} \quad (1)$$

Propagates along the metasurface under a capacitive boundary condition (C_0). The corresponding magnetic field components are:

$$H_i = \frac{E_0}{\eta_0} \left(\frac{\sqrt{3}}{2} \hat{x} - \frac{1}{2} \hat{z} \right) \sin(10\pi x + 10\pi\sqrt{3}z - 6.0 \times 10^{10} \pi t) \quad (2)$$

For a transverse electric (TE) surface mode (β, ω_0) traveling along a capacitive interface with capacitance C_1 , the corresponding electric and magnetic fields are expressed in Equation (3). At $t = 7$ ns, activating the voltage-controlled switch between the copper plates modifies the boundary condition, turning it from a capacitive interface into an inductive one with an associated inductance L . This sudden change permits only transverse magnetic (TM) surface modes to propagate

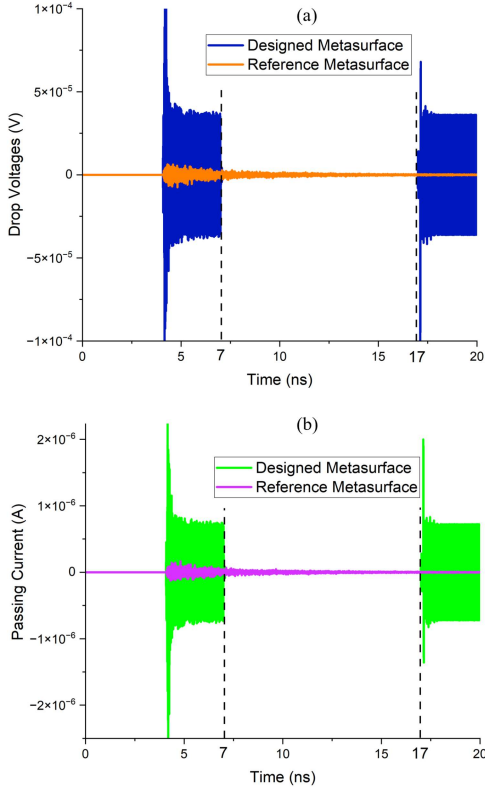


Fig. 4: A comparative analysis of voltage and current absorption in the resistive load of the metasurface is illustrated. In this configuration, the source is positioned on one edge, while the resistive load is placed at the opposite edge of the structure. Figures (a) and (b) demonstrate that the resistive load effectively absorbs significant voltage and current throughout the 20 ns simulation. However, absorption is deliberately suppressed between 7 ns and 17 ns to achieve controlled operation of the designed metasurface. In contrast, for the reference copper metasurface, no control mechanism is introduced, as the resistive load receiving negligible voltage and current over the entire simulation period.

at nonzero frequencies, effectively replacing the TE mode. Nevertheless, a TE mode may still exist on an inductive interface in free space when its frequency tends toward zero ($\omega_1 = 0$). Immediately after the switching event ($t = 7^+$), the z -component of the magnetic field is described as follows:

$$H_z^{(t=7^+)} = H_0 e^{-j\beta x - \alpha_1 z} \hat{z},$$

where

$$H_0 = \frac{-E_0 \beta}{\omega_0 \mu_0},$$

ensuring conservation of the magnetic field's z -component after the transition. Since $\omega_1 = 0$, the time-harmonic term

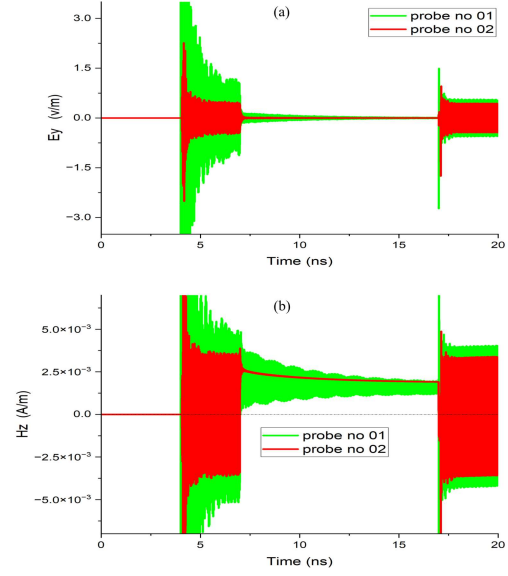


Fig. 5: This section presents the actual readings from two probes positioned at two distinct points. Probe1 is located near the incident side of the metasurface, while Probe2 is placed at the farthest position. The designed structure effectively guides the y -polarized electromagnetic waves, encompassing both the magnetic field (H_z) and the electric field (E_y). Figures (a) and (b) illustrate the zero and frozen field distributions of the electric and magnetic components, respectively, within the time window from 7ns to 17ns.

$e^{j\omega_1 t}$ disappears, and the remaining field components simplify as:

$$H_y^{(t=7^+)} = 0, \quad H_x^{(t=7^+)} = jH_0 e^{-j\beta x - \alpha_1 z} \hat{x},$$

$$E_x^{(t=7^+)} = E_y^{(t=7^+)} = E_z^{(t=7^+)} = 0.$$

This demonstrates a static magnetic field that varies spatially along the x and z directions, whereas the electric field is zero in all directions. The conversion from a capacitive to an inductive boundary, controlled by a voltage-operated switch, is further depicted through the two-dimensional distributions of the electric and magnetic fields in Figures 6(a–c). This type of inductive boundary, often described as a “frozen eigenmode,” satisfies the following relation:

$$j\omega_1 L_1 \cdot (\mathbf{n} \times H_x^{(t=7^+)}) = E_y^{(t=7^+)} = 0.$$

Probing various positions along the metasurface, as shown in Fig. 6(d), reveals the instantaneous behavior of the electric (E_y) and magnetic (H_z) fields at different locations. Probe 1, situated near the source, exhibits a sudden drop of the electric field to zero, while the magnetic field remains nearly frozen with only minor oscillations. This effect becomes more pronounced at higher frequencies, such as 30 GHz, confirming

the expected vanishing of the electric field in the transition region. Correspondingly, the voltage and current across the

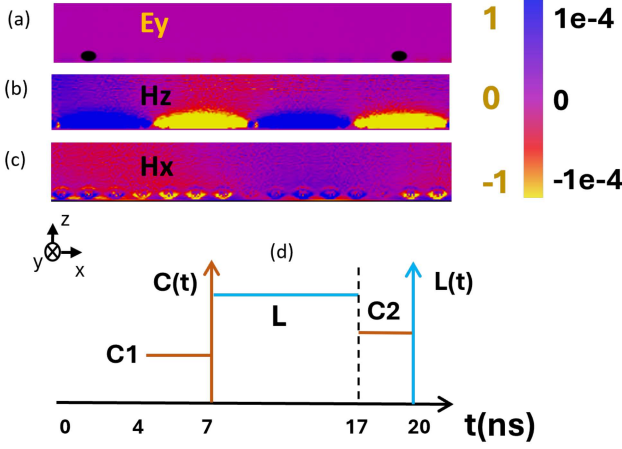


Fig. 6: The first three figures present the two-dimensional distributions of the electric and magnetic fields along the metasurface during the freezing interval, $7 < t < 17$ ns. Figure (a) demonstrates that the electric field is completely suppressed during the freezing state. In Figures (b) and (c), the magnetic field components H_x and H_z exhibit a 90° phase difference, in agreement with theoretical predictions. Figure (d) shows the temporal evolution of the boundary properties over the entire simulation.

resistive load also diminish to zero, as shown in Figs. 4(a) and 4(b). The inductance is a key factor controlling the duration of this transient electric field disappearance at $t = 7$ ns, with both the transient strength and timing being proportional to the inductance value. This effect can be theoretically analyzed using the Laplace transform approach [24]. At $t = 17$ ns, the boundary condition returns to a capacitive state with a new surface capacitance $C2$, determined from the eigenvalue equation of the surface mode frequency ω_2 . Consequently, the electric field reappears via reflection and transmission, allowing wave propagation to resume after the second transition. According to Maxwell's equations in free space, the time derivatives of the electric flux density ($\mathbf{D}$) and magnetic flux density ($\mathbf{B}$) correspond to the curl of the magnetic field ($\mathbf{H}$) and electric field ($\mathbf{E}$), respectively. In addition, the continuity of these field vectors must be preserved during abrupt changes in the surface impedance. Therefore,

$$\mathbf{D}(t = 7^-) = \mathbf{D}(t = 7^+), \quad \mathbf{B}(t = 7^-) = \mathbf{B}(t = 7^+). \quad (3)$$

By employing Equation (4), at $t = 17$, i.e.,

$$E_{ry}^{(t=17^+)} + E_{ty}^{(t=17^+)} = E_y^{(t=17^-)} = 0,$$

$$H_{rz}^{(t=17^+)} + H_{tz}^{(t=17^+)} = H_z^{(t=17^-)},$$

the reflection and transmission coefficients at $t = 17$ ns can be evaluated after the second boundary change, as outlined

in [25]. According to the duality principle, converting the boundary from inductive to capacitive causes the TM-polarized electric field to become a non-propagating, static surface wave. Reversing this boundary shift reinstates the TM surface wave, allowing the electric field to propagate once again. Comparable effects have been observed in cases such as sudden plasma generation [26] and magnetized plasmas subjected to rapidly changing bias fields [27]. Between 7–17 ns, the voltage drops to zero because the electric field temporarily vanishes while the magnetic field stays constant. Along the probes, the field is strongest near the metasurface edges and weakens with distance. This field-freezing effect allows the magnetic field to persist as the electric field disappears, controlling energy localization and wave propagation.

In our study, we demonstrate that the flow of current to the load can be temporarily interrupted, bringing the voltage drop across the load to zero. Following the second boundary transition, wave propagation resumes, restoring both the current and voltage. This approach enables precise regulation of power absorption by the load, which is positioned at the far edge of the metasurface relative to the antenna radiating electromagnetic waves into free space.

V. CONCLUSION

This work explores how temporal boundaries in metasurfaces can serve as an active mechanism to control the voltage and current across a resistive load. Within the framework of temporal discontinuities, which impose momentum conservation and enable frequency conversion, we relate established theory to artificial metasurfaces. Prior studies indicate that time-varying materials can introduce non reciprocity, frequency mixing, and dynamic field confinement, while spatiotemporal modulated metasurfaces support real-time wave manipulation. To illustrate this behavior, a full-wave simulation was performed in which a horn antenna illuminated a reconfigurable metasurface with Y-polarized waves at an incidence angle of 30° . The surface rapidly switched between capacitive and inductive states through controlled voltage signals, thereby creating temporal boundaries. Using exponential signals, the abrupt impedance transitions were shown to suppress or restore the load voltages and currents effectively, while PEC blocks ensured efficient coupling between the source and the load. These results demonstrate that sudden boundary transitions enable precise and tunable control of energy, surpassing the limitations of static metallic surfaces. The findings confirm recent theoretical predictions and highlight practical pathways toward adaptive communication systems, programmable absorbers, and active sensing. Future work may include experimental validation, integration with ultrafast switching components, and extension of temporal control to multifunctional metasurfaces for real-time field manipulation.

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